Connectors

J2
GPIO EXPANSION A
3V3 GND FPGA_RX FPGA_TX GPIO_A0 GPIO_A1 GPIO_A2 GPIO_A13 GPIO_A14

J3
GPIO EXPANSION B
3V3 GND FLASH_CS FLASH_MISO FLASH_MOSI FLASH_SCK FPGA_RESET_N GPIO_B6 GPIO_B9 GPIO_B10 GPIO_B11

J4
SPI PROGRAM
3V3 GND FLASH_CS FLASH_SCK FLASH_MOSI FLASH_MISO

U1 ICE40UP5K-SG48I

U1A
ICE40UP5K-SG48I
VCCIO_0
IOT_36b IOT_37a IOT_38b IOT_39a IOT_41a IOT_42b IOT_43a IOT_44b IOT_45a_G1 IOT_46b_G0 IOT_48b IOT_49a IOT_50b IOT_51a RGB0 RGB1 RGB2

U1B
ICE40UP5K-SG48I
SPI_VCCIO1
CRESET CDONE
IOB_13b IOB_16a IOB_18a IOB_20a IOB_22a IOB_23b IOB_24a IOB_25b_G3 IOB_29b IOB_31b IOB_32a_SPI_SO IOB_33b_SPI_SI IOB_34a_SPI_SCK IOB_35b_SPI_SS

U1C
ICE40UP5K-SG48I
VCCIO_2
IOB_0a IOB_2a IOB_3b_G6 IOB_4a IOB_5b IOB_6a IOB_8a IOB_9b

U1D
ICE40UP5K-SG48I
VPP_2V5 VCC VCCPLL GND

C2 10uF 10V
C3 1uF 50V
C8 100nF 50V
C18 100nF 50V
C16 10uF 10V
C6 10uF 10V
C14 100nF 50V
C11 100nF 50V
C7 100nF 50V

R5 10k
R6 10k
R7 0R
R8 10k

USB — U2 CH340C

U2
CH340C
R232 VCC V3 UD+ UD− TXD RXD CTS DSR RI DCD DTR RTS GND

C9 100nF 50V
C10 100nF 50V
C4 10uF 10V

Connectors — J1 TYPE-C 16PIN 2MD(073)

J1
TYPE-C 16PIN 2MD(073)
VBUS CC1 CC2 D− D+ SBU1 SBU2 SHIELD GND A1 SH

R1 5.1k
R2 5.1k

RV32I is a SystemVerilog soft CPU implemented inside FPGA U1, not a purchased PCB part. U3 stores its FPGA configuration; U2 provides UART debug; J2/J3 expose GPIO. CPU RTL and firmware must be developed and validated separately.

U3 W25Q32JVSSIQ

U3
W25Q32JVSSIQ
CS DO(IO1) WP#(IO2) GND VCC HOLD#orRESET#(IO3) CLK DI(IO0)

R3 10k
R4 10k

Linear regulator — U4 AP2112K-3.3TRG1

U4
AP2112K-3.3TRG1
VIN EN GND VOUT

C1 10uF 10V
C5 1uF 50V
C15 1uF 50V

U6 ME6211C25M5G-N

U6
ME6211C25M5G-N
VIN VSS CE VOUT NC

C17 100nF 50V

U5 ME6211C12M5G-N

U5
ME6211C12M5G-N
VIN VSS CE VOUT NC

C12 100nF 50V
C13 100nF 50V

schematic-first design.json
heypcb.ai
Sheet: /
File: i-am-building-a-rv32i-fpga-soc-development-board.kicad_sch

Title: risc-v

Size: C	Date: 2026-10-13	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1